Paramount Diagnostic Lab

Carbohydrate Antigen (CA) 19-9

Test Details

METHODOLOGY

Electrochemiluminescence immunoassay (ECLIA)

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Monitor gastrointestinal, pancreatic, liver, and colorectal malignancies

Special Instructions

Values obtained with different assay methodologies should not be used interchangeably in serial testing. It is recommended that only one assay method be used consistently to monitor a patient's course of therapy. This procedure does not provide serial monitoring; it is intended for one-time use only. If serial monitoring required, please use the serial monitoring number 480053 (/tests/480053/carbohydrate-antigen-ca-19-9-serial-monitor) to order.

This test may exhibit interference when sample is collected from a person who is consuming a supplement with a high dose of biotin (also termed as vitamin B7 or B8, vitamin H, or coenzyme R). It is recommended to ask all patients who may be indicated for this test about biotin supplementation. Patients should be cautioned to stop biotin consumption at least 72 hours prior to the collection of a sample.

Limitations

The measured CA 19-9 value of a patient's sample can vary depending on the testing procedure used. CA 19-9 values determined on patient samples by different testing procedures cannot be directly compared with one another and could be the cause of erroneous medical interpretations.

The determination of CA 19-9 cannot be used for the early detection of pancreatic carcinoma.[1]

Three percent to 7% of the population have the Lewis a-negative/b-negative blood group configuration and are unable to express the mucin with the reactive determinant CA 19-9. Patients known to be genotypically negative for Lewis blood group antigens will be unable to produce the CA 19-9 antigen even in the presence of malignant tissue. Phenotyping for the presence of the Lewis blood group antigen may be insufficient to detect true Lewis antigen-negative individuals. Even patients who are genotype positive for the Lewis antigen may produce varying levels of CA 19-9 as the result of gene dosage effect. This must be taken into account when interpreting the findings.[2]

As the mucin is excreted exclusively via the liver, even slight cholestasis can lead to clearly elevated CA 19-9 serum levels in some cases.

As with all tests containing monoclonal mouse antibodies, erroneous findings may be obtained from samples taken from patients who have been treated with monoclonal mouse antibodies or who have received them for diagnostic purposes.[1] In rare cases, interference due to extremely high titers of antibodies to streptavidin and ruthenium can occur.[1] The test contains additives, which minimize these effects.

For diagnostic purposes, the results should always be assessed in conjunction with the patient's medical history, clinical examination, and other findings.

Custom Additional Information

This assay intended for the in vitro quantitative determination of CA 19-9 tumor-associated antigen in human serum and plasma.[1] The assay is indicated for the serial measurement of CA 19-9 to aid in the management of patients diagnosed with cancers of the exocrine pancreas. The test is useful as an aid in the monitoring of disease status in those patients having confirmed pancreatic cancer who have levels of CA 19-9 at some point in their disease process exceeding the median concentration determined for the apparently healthy cohort.

The CA 19-9 values measured are defined by the use of the monoclonal antibody 1116-NS-19-9. The 1116-NS-19-9-reactive determinants on a glycolipid having a molecular weight of approximately 10,000 daltons are measured. This mucin corresponds to a hapten of Lewis-a blood group determinants and is a component of a number of mucous membrane cells.[3,4]

Mucin occurs in fetal gastric, intestinal, and pancreatic epithelia. Low concentrations can also be found in adult tissue in the liver, lungs, and pancreas.[2,5]

CA 19-9 assay values can assist in the differential diagnosis and monitoring of patients with pancreatic carcinoma (sensitivity 70% to 87%).[6-12] There is no correlation between tumor mass and the CA 19-9 assay values; however, patients with CA 19-9 serum levels >10,000 units/mL almost always have distal metastasis.[2]

Specimen Requirements

Specimen

Serum

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Gel-barrier tube (preferred) or red-top tube

Collection Instructions

If a red-top tube is used, transfer separated serum to a plastic transport tube.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

0−35 units/mL

Storage Instructions

Room temperature

Causes for Rejection

Citrate plasma specimen; improper labeling

References

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Footnotes

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